

28 Ans: B

Electrons: $I = dQ/dt = d(eN_e) / dt$

$$\square \quad dN_e/dt = I / e = 2.0 \times 10^{-6} / 1.6 \times 10^{-19}$$

Photons: $P = dE/dt = d(N_p \times \text{energy of each photon}) / dt$

$$\square \quad dN_p/dt = P / E = 0.31 \times 10^{-3} / 3.11 \times 1.6 \times 10^{-19}$$

Ratio = 0.020

29 Ans: C